

Self-initiated Project: inDrive

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Role: UI/UX Designer & Product Thinker

Status: High-Fidelity Prototype Completed

About inDrive Next-Gen Platform

This case study looks at how inDrive works as a peer-to-peer (P2P) fare negotiation service and how we can make the app safer, smarter, and easier for everyone to use. We are suggesting ways to improve the design so it is fully accessible (WCAG compliant), using smarter AI and MIP engines to help you find the best fair-price matches, integrating seamless auto-deduct payments, and adding critical biometric security features to protect both drivers and customers from account misuse.

The Core Problems

I didn't just guess what was wrong; I mapped out the information architecture, ran operational field tests, and broke down the user journeys. Through this, four massive friction points became glaringly obvious:

- **The Identity Gap (Security & Trust)**

Right now, the platform has zero real-time facial verification for drivers. This is a critical vulnerability. It leaves the door wide open for account sharing or hijacking, putting passengers at serious risk and slowly eroding trust in the platform.

- Cognitive Overload on the Road

The driver dashboard is, frankly, chaotic. Ride-hailing and courier orders are dumped into a single, unorganized feed. Drivers are forced to mentally filter task types and manually punch in prices while trying to focus on the road. It's not just bad UX; it's unsafe.

- Bidding & Payment Friction

Typing out fares manually while dealing with cash or manual bank transfers turns what should be a quick bid into a slow, stressful negotiation. Without AI to anchor the price context, and without an auto-deduct system, we end up with delayed transactions, the awkward "exact change" dilemma, and incredibly high drop-off rates.

- Accessibility & Spatial Failures

Visually impaired users are basically locked out of the app because critical UI elements don't work with screen readers. On top of that, relying solely on raw GPS triangulation causes wild location drift. Drivers end up at the wrong pickup spots, which instantly spikes cancellation rates.

Design Solution & Strategy (Product Strategy)

The design approach focuses on **Simplification through Systemic Logic**, implementing four key UX strategies:

- **Biometric Security & Visual Trust**

I am making AI Facial Verification (Liveness Check) mandatory for drivers before they go online, supplemented with random checks. This guarantees that the person behind the wheel is exactly who the app says they are.

- **Decluttering the Architecture**

I used progressive disclosure—like smart accordions and tooltips—so technical details stay out of the way until the user actually needs them. For drivers, the dashboard is now strictly tab-based (separating City Rides from Courier drops). I also introduced a "Floating Compact Earnings Card" so drivers can scan their income metrics safely without the screen feeling cluttered.

- **Seamless Pre-Authorization**

I baked an Auto-Deduct E-Wallet system directly into the bidding flow. By forcing users to pick their payment method *before* bidding and using pre-authorization, we completely kill the friction of manual transfers, solve the exact change issue, and guarantee drivers get paid.

- **AI-Assisted Decisions**

I replaced manual typing with a Dynamic Bidding Slider. It's powered by a Mixed Integer Programming (MIP) engine and a GenAI RAG layer that pulls in real-time context like weather or heavy traffic. It gently guides the user toward a fair price without taking away their control.

Feature & Functionality Requirements

Below is the breakdown of the product requirements implemented in the user interface:

A. Onboarding & Security

Visual onboarding introduces the modern inDrive identity, highlighting fair pricing, seamless payments, and enhanced safety protocols.

- **Biometric Authorization (Facial Verification)**

A dedicated, high-friction security screen requiring drivers to perform a real-time liveness check (e.g., "blink" or "turn head") using device camera access before starting their shift.

- **Sign In / Sign Up Flow**

Simplified authentication flows with clear input validation to minimize errors.

B. Core Navigation

A tabbed interface strictly separating service contexts. Features a "Radius Control Slider" allowing drivers to limit incoming request distances.

- **Floating Compact Earnings Card**

A glanceable, collapsible UI element pinned at the top of the driver's screen displaying "Today's Earnings" and "Driver Balance," expandable for instant withdrawal actions.

- **Unified Customer Home Screen**

A personalized map layout prioritizing the destination search bar, quick "Saved Places" chips, and a high-accuracy "Absolute Pin" with a GPS confidence score overlay.

C. The Intelligence Hub (Bidding & Matching)

A mandatory step where customers select Cash or Cashless (OVO/GoPay/Dana) *before* bidding. Cashless selection activates a "Priority Match" badge. Cash selection triggers an "Exact Change Required" input field.

- **Dynamic Bidding Slider & RAG Tooltips**

An interactive, color-coded slider (Green/Amber/Red) proposing fair fares based on real-time external data. Contextual AI tooltips (e.g., "High demand—suggested +10%") provide pricing transparency.

- **Pre-Authorization & Instant Refund UI**

System holds the balance securely. If an order is canceled, a full-width, highly visible confirmation card guarantees the customer that their money has been instantly refunded, reducing panic and support tickets.

- **Business Receipt Mode & History**

Detailed trip history featuring driver identity, route map, and downloadable PDF invoices for corporate reimbursement needs.

D. System States & Accessibility

Transparent system permission confirmation interfaces, explicitly explaining the need for Camera Access (for Facial Verification security) and Precise Location Access (for Mapbox integration and MIP routing).

- **Loading & Negotiation States**

Visual feedback (pulsing indicators) provided while the MIP engine calculates the best driver-order assignment, maintaining users' psychological comfort during the matching wait time.

- **Accessibility (A11y) Compliance Layer**

All interactive elements are built with semantic labels, ensuring full compatibility with TalkBack and VoiceOver screen readers.

How I Measure Success

Even though this is a high-fidelity prototype, I've designed it with clear goals in mind to make sure the app really works safely and efficiently for people. Here is how I will know we are on the right track:

Enhanced Security & Trust (Zero Hijacking)

The primary metric for the new biometric system is reducing driver impersonation reports to absolute zero, significantly increasing customer trust scores and post-trip ratings.

Speed and Ease of Use

The goal is for customers to complete the bidding and matching process in under 30 seconds (down from 87 seconds), and for drivers to cognitively process incoming order cards in under 4 seconds.

Frictionless Financial Operations

By introducing seamless auto-deduct and pre-authorization, I expect a massive drop in canceled rides caused by payment disputes, and a near 100% elimination of "missing change" complaints.

Happier, Safer Users

Ultimately, I want to see a significant jump in retention rates for both drivers (who feel safer and less distracted) and customers (who experience fair, transparent, and instantly verifiable mobility services).